VibesYeti

A Subwoofer for Your Heart

SCHOOL OF DESIGN
创新创意设计学院

Design Background

Accelerating urbanization and shifting family structures in China and Southeast Asia have increased parent-child separation. Many rural children are "left behind" as parents migrate for work, while urban families face fragmentation from nontraditional schedules. This separation can challenge caregiving and negatively impact children's emotional well-being, leading to loneliness and anxiety.

Parenting practices are changing; traditional co-sleeping is being replaced by an emphasis on independence, which may heighten emotional insecurity. Children with special needs, such as those on the autism spectrum or hearing impairments, face even greater emotional disconnection due to physical separation.

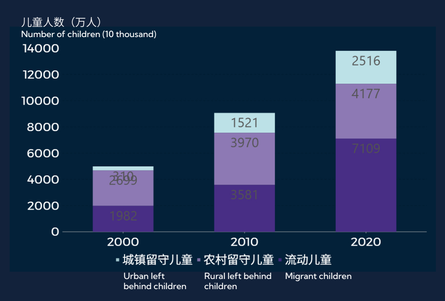
To address these challenges, we explore how interaction design can help maintain emotional connections across distances using technologies like tactile interfaces and remote feedback to provide warmth and reassurance to children in their parents' absence.

设计背景

随着中国及东南亚地区城市化进程的加速以及家庭结构的持续变迁，亲子分离现象日益突出。大量农村儿童因父母外出务工而成为“留守儿童”，而城市家庭则因非传统作息安排面临家庭关系的碎片化。这种空间与时间上的分离对养育实践提出了严峻挑战，并对儿童的情感健康产生负面影响，表现为孤独感与焦虑情绪的上升。

与此同时，育儿观念也在发生转变：传统的亲密共睡方式正在逐步被强调儿童独立性的现代养育理念所取代，这在一定程度上可能加剧儿童的情感不安全感。对于特殊需求儿童（如自闭症谱系障碍或听力障碍儿童）而言，身体上的分离更容易导致情感连接的进一步削弱。

为应对上述问题，本研究探讨了互动设计在维系远距离亲子情感联结中的潜力，尤其关注触觉界面与远程反馈等技术手段，如何在父母无法陪伴时，为儿童提供温暖与情感安抚，从而缓解其情绪压力并提升安全感。



Number of children affected by migration, 2000, 2010 and 2020

受人口流动影响的儿童人数，2000 年、2010 年和 2020 年

Design Questions

Can we design a tactile-mediated companionship system that



...addresses emotional needs with haptic experiences?

... 通过触觉体验满足情感需求?



...evolves with children's different developmental stages without fostering technological overreliance?

... 在不培养技术过度依赖的情况下，随儿童的成长而发展?



...serves as an ethical bridge rather than replacement for human bonds?

... 作为一个道德的桥梁，而不是取代人类的纽带?

设计问题

我们是否可以设计一个触觉介导的陪伴系统

Design Research Methods

Through literature, user research, character development, prototyping.

设计研究方法

通过文献、用户研究、角色开发、原型制作。

Product Description

VibesYeti is a plush toy designed for children aged 2 to 14, featuring haptic output capabilities. It can be cuddled and played with like any toy, while also offering low-frequency vibrations that provide emotional comfort and a sense of connection between parents and children.

It operates in two modes:

1. Online/remote: Wirelessly connects to a parent or caregiver's microphone or sensor, streaming real-time sounds like singing, humming, and heartbeat to create a sense of presence.

2. Offline/recorded: Functions as a standalone device that plays pre-recorded vibrations, including soothing sounds like humming and purring.



Offline 离线



Online 在线

产品描述

VibesYeti 是一款面向 2 至 14 岁儿童设计的毛绒玩具，具备触觉反馈功能。它既可作为普通玩具供儿童拥抱与玩耍，又能通过低频振动传递情感慰藉，在父母与子女之间营造出连接感与陪伴感。

该产品具备两种运行模式：

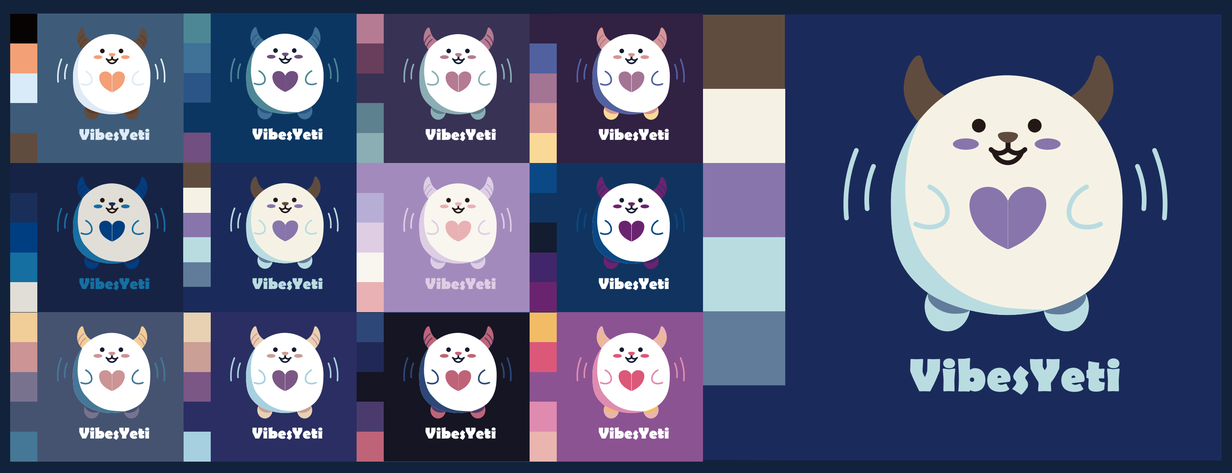
在线／远程模式：可无线连接至父母或照护者的麦克风或传感器，实时传输如歌唱、哼唱、心跳等声音信号，以增强“在场感”与亲密互动体验。

离线／预录模式：作为独立设备运行，播放预先录制的振动信号，包括哼唱、咕噜声等舒缓音效，从而为儿童提供持续的情感安抚。

Design Development

Logo and Color Scheme

Keywords: **tranquility, late night, security**...



设计发展

标志和配色方案

关键词：**“宁静”、“深夜”、“安全”**...

Character Development

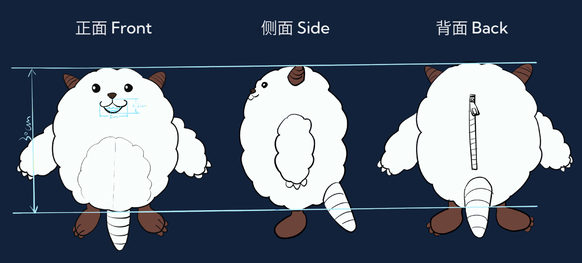
Core rationale for choosing Yeti as the design silhouette:

Safety & Comfort
The Yeti's smooth, fur-covered shape gives a sense of security. Its design, with no sharp edges, feels like a warm embrace, making children feel safe.

Childhood Fascination
As a mythical creature, the Yeti sparks kids' curiosity and imagination, acting like a special protector. Its look is a mix of real and abstract, making it relatable for children.

Inclusiveness
The Yeti features friendly traits, like round eyes and a smile, making it appealing to all kids and crossing traditional gender lines.





角色开发

选择雪怪作为设计轮廓的核心理由：

安全与舒适
雪怪圆润的毛绒外型带来油然而生的安全感，无棱角的设计宛如温暖怀抱，让孩子倍感安全。

童年吸引力
作为神话生物，雪怪能激发孩子的好奇心与想象力，如同一位特殊保护者。其造型融合现实与抽象特质，让孩子产生共鸣。

包容性设计
雪人拥有圆眼睛、微笑等友好特质，对所有孩子都具有吸引力，打破了传统的性别界限。

User Scenario

The storyboard highlights how children feel uneasy when their parents aren't with them at bedtime, but can sleep peacefully after using VibesYeti.

用户场景

故事板展示了当父母不在身边时，儿童在入睡前感到不安的情境；而在使用 VibesYeti 后，他们能够安然入睡。



